Lesson 10: Play It Again

Unreal Engine (optional advanced track)

VIDEO GAME CREATION

This week: when the round ends, the game actually stops - and pressing R brings it back to a genuinely fresh start.

Open your own [YourName]Game project from Lesson 9 - don't start a new one.

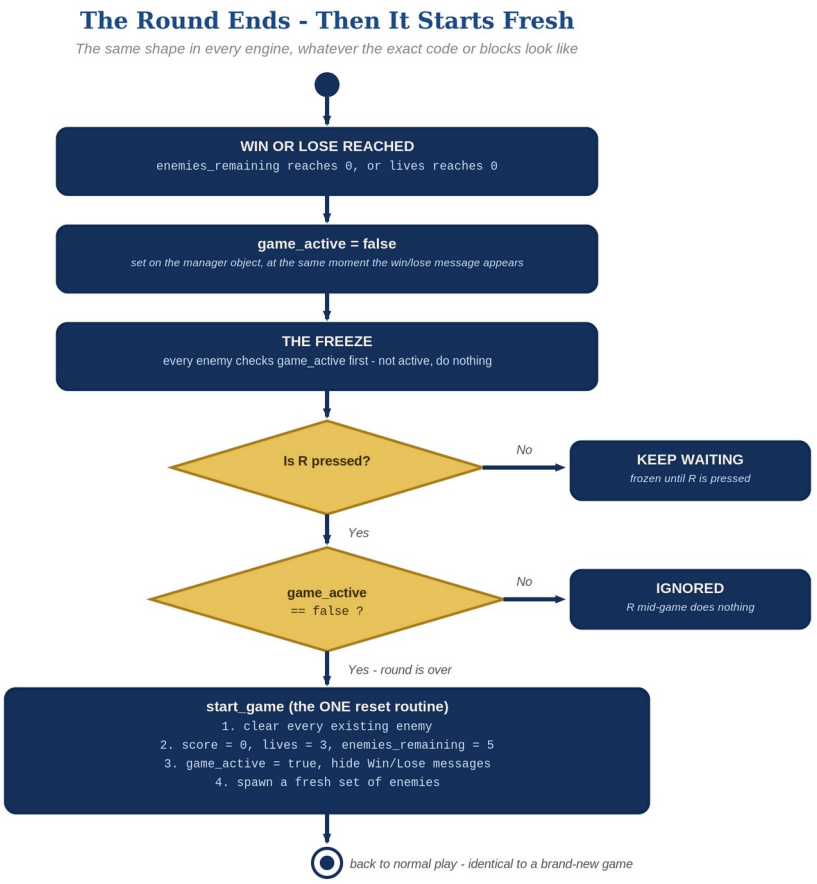
Tip: no new assets this week - this is a logic lesson, and it reuses your very first setup code from Lesson 5.

Setup

1. Open your [YourName]Game project from Lesson 9.
2. In WBP_HUD's Designer view, add a third Text Block named StatusText, positioned centrally, initially empty.
3. Open BP_GameManager and add a new Boolean variable GameActive, default value true - it joins Score, Lives, and EnemiesRemaining as shared game state.
4. In the Content Browser, click Add (+) > Input > Input Action, name it IA_Restart. Leave its Value Type as the default (Digital bool), same as your other Input Actions.
5. Open IMC_Default (the Input Mapping Context from Lesson 2) and click + Add Mapping. Choose IA_Restart, then set its Key to R. Since IMC_Default is already applied to the player, this new mapping is live immediately.

Level 1 - Minimal (MVP)

When the round ends (win or lose), the game freezes - score and lives stop changing, and further enemy hits/escapes do nothing. Pressing R resets everything and starts a new round.



When the round ends, everything freezes - and R, pressed only while the round is over, runs the one reset routine. The same shape in every engine.

6. In WBP_HUD's Event Graph, add a Custom Event SetStatus(StatusText: String) that sets the StatusText Text Block's text from the input (code below).
7. In BP_Enemy's Event Graph, add a Branch at the very start of Event Tick that reads GameActive from BP_GameManager (use Get Actor of Class(BP_GameManager), then Get GameActive). Wrap all existing movement logic inside the True case, so nothing moves once the round ends.
8. In BP_GameManager, build a new Custom Event ResetGame: destroy all existing BP_Enemy actors, reset Score/Lives/EnemiesRemaining/GameActive, call SetStatus with an empty string, call UpdateHUD, then spawn a fresh set of enemies with the same For Loop your spawn logic already used (code below). BP_GameManager owns the shared game state, so the reset lives here and not in the Level Blueprint.
9. Change BP_GameManager's Event BeginPlay to create the widget, add it to viewport, set HUDWidget, then call ResetGame instead of spawning enemies directly.
10. In HandleEnemyDestroyed's win Branch and HandleEnemyEscaped's lose Branch, when the round ends, also set GameActive = false and call SetStatus with the matching restart-hint message.

11. In BP_GameManager's Event Graph, right-click and search for IA_Restart to add an Enhanced Input Action Event node: on its Started pin, Branch on !GameActive -> True: Call ResetGame. (If BP_GameManager does not receive input in your project, enable Auto Receive Input on it in Class Defaults, or add the node to your player Blueprint and call ResetGame on BP_GameManager from there.)
12. Click Compile, then Play, play to a win or loss, confirm movement freezes, then press R and confirm everything resets and play resumes.

Highlighted lines are new this week; everything else is code you already have.

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WBP_HUD's Event Graph (new):
Custom Event SetStatus(StatusText: String) ->
  Set Text(StatusTextBlock, StatusText)

BP_Enemy's Event Graph (revised):
Event Tick ->
  Branch(Get GameActive from Get Actor of Class(BP_GameManager)) -> True:
    Add Actor World Offset(Delta = (0, -EnemySpeed * Delta Seconds, 0))
    Branch(GetActorLocation().Y < -400) -> True: Call HandleEnemyEscaped
on BP_GameManager -> Destroy Actor(Self)

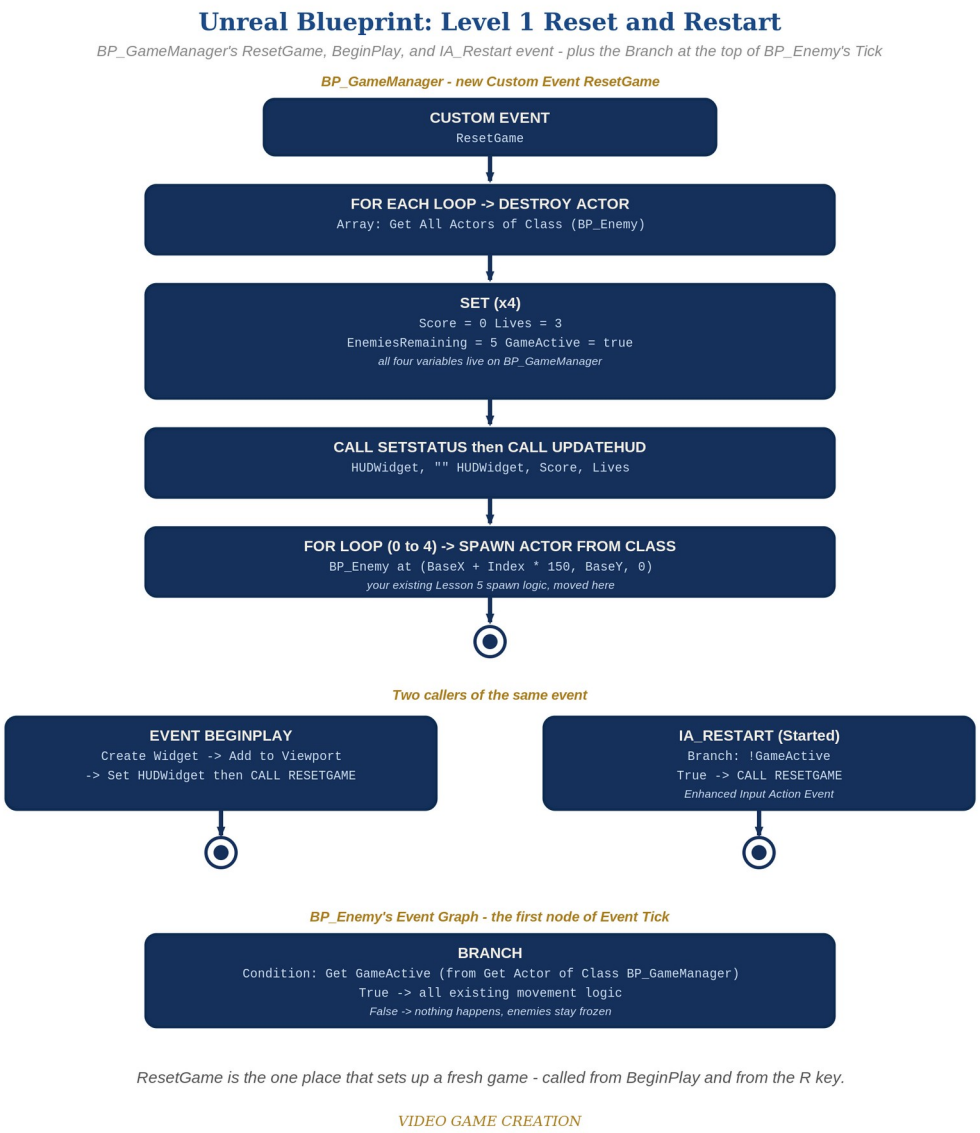
BP_GameManager (variables: Score, Lives, EnemiesRemaining, HUDWidget,
GameActive):
Custom Event ResetGame ->
  For Each (Get All Actors of Class(BP_Enemy)) -> Destroy Actor
  Score = 0
  Lives = 3
  EnemiesRemaining = 5
  GameActive = true
  Call SetStatus(HUDWidget, "")
  Call UpdateHUD(HUDWidget, Score, Lives)
  For Loop (First Index 0, Last Index 4) -> Spawn Actor from
Class(BP_Enemy) at (BaseX + Index * 150, BaseY, 0) -> Bind Event to
OnEnemyDestroyed / OnEnemyEscaped

Event BeginPlay ->
  Create Widget(WBP_HUD) -> Add to Viewport -> Set HUDWidget
  Call ResetGame

Custom Event HandleEnemyDestroyed ->
  EnemiesRemaining = EnemiesRemaining - 1
  Score = Score + 10
  Call UpdateHUD(HUDWidget, Score, Lives)
  Branch(EnemiesRemaining <= 0) -> True:
    GameActive = false
    Call SetStatus(HUDWidget, "You Win! Press R to Restart")

Custom Event HandleEnemyEscaped ->
  Lives = Lives - 1
  Call UpdateHUD(HUDWidget, Score, Lives)
  Branch(Lives <= 0) -> True:
    GameActive = false
    Call SetStatus(HUDWidget, "Game Over - Press R to Restart")

Enhanced Input Action Event (IA_Restart), Started ->
  Branch(!GameActive) -> True: Call ResetGame
```



BP_GameManager's ResetGame, BeginPlay and IA_Restart event, plus the Branch at the top of BP_Enemy's Tick - the node-for-node equivalent of the Level 1 code.

Alternate Path: C++ (code, optional advanced track)

The same reusable-reset pattern works in C++ instead of Blueprint nodes.

13. Add a ResetGame() function to your game manager class that iterates TActorIterator<ACppEnemy> to destroy existing enemies, resets Score/Lives/EnemiesRemaining/bGameActive, and calls your existing spawn function.
14. In ACppEnemy::Tick(), add an early return (if (!bGameActive) return;) at the very top, mirroring the Blueprint path's Branch gate.

This mirrors the Blueprint path's ResetGame Custom Event and Tick Branch exactly - same shape, just from code.

Level 2 - Complete (Alpha/Beta)

A restart is a genuinely clean slate - no leftover enemies, no stale score/lives, and it works correctly no matter how many times in a row you play.

- Play to a win, confirm movement freezes, then press R and confirm Score/Lives/EnemiesRemaining are exactly back to 0/3/5 with no leftover BP_Enemy actors.
- Repeat for a loss - same clean reset.
- Press R while the game is still active and confirm NOTHING happens.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: animate StatusText's opacity in WBP_HUD instead of it appearing instantly.
- Structure: notice ResetGame is now the single place that sets up a fresh game - called once from BeginPlay and again from the IA_Restart Enhanced Input Action Event. No spawn logic is duplicated, and all shared state still lives on BP_GameManager.

If Something's Wrong

- Restart does nothing when you press R: confirm IA_Restart actually has an R mapping inside IMC_Default, that the Enhanced Input Action Event node is wired directly in the Event Graph (not nested inside another event), that BP_GameManager is set to receive input, and that GameActive is actually false at that point.
- Restart leaves old enemies on screen: confirm "Get All Actors of Class(BP_Enemy)" -> Destroy Actor runs BEFORE the spawn For Loop inside ResetGame, not after.
- Enemies still move after game-over: confirm the Branch(GameActive) wraps ALL of Event Tick's movement logic, not just part of it.
- Get GameActive returns nothing: confirm exactly one BP_GameManager is placed in the level.

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